Homework 7

Mathematical Physics

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PHYS 721

1.6.1 Show that $\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right), \quad -1 < x < 1.$

Goal : Find an expansion for $\ln\left(\frac{1+x}{1-x}\right)$

Solution

Recall the sum of a Geometric Series :

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \qquad |x| < 1$$

Integrating w.r. to x yields :

$$-\ln|1-x| = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$$

Replacing x with $-x$

$$\ln|1+x| = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$$

Using both expansions

$$\ln\left(\frac{1+x}{1-x}\right) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} + \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$$

$$= \sum_{n=1}^{\infty} \underbrace{\left[(-1)^{n-1} + 1\right)]}_{\begin{array}{l}=2 \quad n \text{ odd} \\ =0 \quad n \text{ even}\end{array}} \frac{x^n}{n}$$

$$= 2\sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right)$$

12.1.1 Starting from the Rodrigues formula in Table 12.1 for the Hermite polynomials H_n, derive the generating function for the H_n given in that table.

Goal: Derive the generating function for the Hermite polynomials H_n.

Solution

Recall the Hermite ODE:

$$y'' - 2xy' + 2ny = 0$$

To make this ODE self-adjoint, we multiply by weight $f^n\, w(x)$:

where $w(x) = P_0(x)^{-1} \exp\left\{\int \frac{P_1}{P_0}\, dx\right\}$

$= \exp\{\int -2x\, dx\}$

$= e^{-x^2}$

We arrive now to the Rodriguez Formula for H_n:

$$H_n(x) = \frac{1}{w(x)} \cdot \frac{d^n}{dx^n}[wp^n]$$

$$= \underbrace{(-1)^n}_{\text{cons.}}\, e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$$

The generating function is defined as:

$$g(x,t) = \sum_{t=0}^{\infty} H_n(x)\, \frac{t^n}{n!}$$

Writing $H_n(x)$ as a Schlaefli Integral:

$$H_n(x) = (-1)^n e^{x^2} \cdot \frac{n!}{2\pi i} \oint \frac{e^{-z^2}}{(z-x)^{n+1}}\, dz$$

Plugging back H_n in $g(x,t)$ yields:

$$g(x,t) = e^{x^2} \cdot \frac{1}{2\pi i} \sum_{t=0}^{\infty} (-1)^n t^n \oint \frac{e^{-z^2}}{(z-x)^{n+1}}\, dz$$

Flipping the order of Σ and $\oint$:

$$g(x,t) = e^{x^2} \cdot \frac{1}{2\pi i} \oint \sum_{t=0}^{\infty} (-1)^n t^n \frac{e^{-z^2}}{(z-x)^{n+1}} \, dz$$

$$= \frac{1}{2\pi i} e^{x^2} \oint \frac{e^{-z^2}}{z-x} \sum_{t=0}^{\infty} \left(\frac{-t}{z-x} \right)^n dz$$

Rewriting

$$= \frac{1}{2\pi i} e^{x^2} \oint \frac{e^{-z^2}}{z-x+t} \, dz$$

Sum of Geometric Series

Simple pole at $z = x-t$

$$= \frac{1}{2\pi i} e^{x^2} \cdot 2\pi i \operatorname*{Res}_{z=x-t}$$

$$= e^{x^2} \cdot e^{-(x-t)^2}$$

$$= e^{t(2x-t)} \quad \text{OR} \quad e^{-t^2+2xt}$$

▨

13.1.1 Derive the recurrence relations

$$\Gamma(z+1) = z\Gamma(z)$$

from the Euler integral, Eq. (13.5),

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt.$$

Goal: Derive $\Gamma(z+1) = z\Gamma(z)$

Solution

$$\Gamma(z+1) = \int_0^\infty \underbrace{e^{-t}}_{} \underbrace{t^z}_{u} \underbrace{dt}_{dv}$$

$$= t^z \cdot -e^{-t} \Big|_0^\infty - \int_0^\infty -e^{-t} \cdot z t^{z-1} dt$$

$$= z \int_0^\infty e^{-t} t^{z-1} dt$$

$$= z\Gamma(z)$$

13.1.6 Show that

$$\int_0^\infty e^{-x^4} dx = \Gamma\left(\frac{5}{4}\right).$$

Goal: Show that $\int_0^\infty e^{-x^4} dx = \Gamma(5/4)$

Integration by Substitution

$t = x^4$

$dt = 4x^3 dx$

$dt = 4t^{3/4} dx$

Solution

$$\int_0^\infty e^{-x^4} dx = \int_0^\infty e^{-t} \cdot \frac{1}{4} t^{-3/4} dt$$

$$= \frac{1}{4}\Gamma(1/4) \qquad \text{def of } \Gamma \text{ f}^\wedge$$

$$= \frac{1}{4} \frac{\Gamma(5/4)}{1/4} \qquad \frac{\Gamma(z+1)}{z} = \Gamma(z)$$

$$= \Gamma(5/4)$$

13.1.7 Show that

$$\lim_{x\to 0}\frac{\Gamma(ax)}{\Gamma(x)}=\frac{1}{a}.$$

Goal : $\lim_{x\to 0}\frac{\Gamma(ax)}{\Gamma(x)}=\frac{1}{a}$

Solution

Recall Euler's definition of $\Gamma(x)$

$$\Gamma(x)=\lim_{n\to\infty}\frac{1\cdot 2\cdot 3\cdot\ldots\cdot n}{x\cdot(x+1)\cdot\ldots(x+n)}\,n^{x}\qquad \text{for } x\neq 0,-1,-2,\ldots$$

Then,

$$\lim_{x\to 0}\frac{\Gamma(ax)}{\Gamma(x)}=\lim_{x\to 0}\lim_{n\to\infty}\frac{\cancel{1\cdot 2\cdot 3\cdot\ldots\cdot n}\;n^{ax}}{ax(ax+1)\cdot\ldots(ax+n)}\Big/\frac{\cancel{1\cdot 2\cdot 3\cdot\ldots\cdot n}\;n^{x}}{x\cdot(x+1)\cdot\ldots(x+n)}$$

$$=\lim_{n\to\infty}\lim_{x\to 0}\frac{\cancel{x}\cdot(x+1)\cdot\ldots\cdot(x+n)}{\cancel{ax}(ax+1)\cdot\ldots(ax+n)}\,n^{(a-1)x}$$

$$=\lim_{n\to\infty}\frac{1}{a}\cdot\frac{\cancel{1\cdot 2\cdot\ldots\cdot n}}{\cancel{1\cdot 2\cdot\ldots\cdot n}}\cdot\cancel{n^{0}}$$

$$=\frac{1}{a}$$

* More Elegant proof :

$$\lim_{x\to 0}\frac{\Gamma(ax)}{\Gamma(x)}=\lim_{x\to 0}\frac{\Gamma(ax+1)/ax+1}{\Gamma(x+1)/x+1}\qquad \Gamma(1)=1$$

$$=\lim_{x\to 0}\frac{x+1}{ax+1}=\frac{1}{a}$$

13.1.10 Show that, for integer s,

(a) $$\int_0^\infty x^{2s+1}\exp(-ax^2)dx = \frac{s!}{2a^{s+1}}.$$

(b) $$\int_0^\infty x^{2s}\exp(-ax^2)dx = \frac{\Gamma(s+\frac{1}{2})}{2a^{s+1/2}} = \frac{(2s-1)!!}{2^{s+1}a^s}\sqrt{\frac{\pi}{a}}.$$

Goal: Use Gamma functions to solve integrals

Solution

Recall the Integral formula of $\Gamma(s)$

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\, dt$$

a) Using the formula for $\Gamma(s+1)$

$$\Gamma(s+1) = \int_0^\infty t^s e^{-t}\, dt$$

Integration by Substitution

$t = ax^2$

$dt = 2ax\, dx$

$$= \int_0^\infty (ax^2)^s e^{-ax^2} \cdot 2ax\, dx$$

$$= 2a^{s+1} \int_0^\infty x^{2s+1} e^{-ax^2}\, dx$$

Taking constants to the left-hand side.

$$\int_0^\infty x^{2s+1} e^{-ax^2}\, dx = \frac{\Gamma(s+1)}{2a^{s+1}} = \frac{s!}{2a^{s+1}}$$

$\Gamma(s+1) = s!$ for $s \in \mathbb{Z}$

b) Using the formula for $\Gamma(s+\frac{1}{2})$

$$\Gamma(s+\tfrac{1}{2}) = \int_0^\infty t^{s-\frac{1}{2}} e^{-t}\,dt$$

Integration by Substitution

$$t = ax^2$$

$$dt = 2ax\,dx$$

$$= \int_0^\infty (ax^2)^{s-\frac{1}{2}} e^{-ax^2} \cdot 2ax\,dx$$

$$= 2a^{s+\frac{1}{2}} \int_0^\infty x^{2s} e^{-ax^2}\,dx$$

Taking constants to the left-hand side.

$$\int_0^\infty x^{2s} e^{-ax^2}\,dx = \frac{\Gamma(s+\frac{1}{2})}{2a^{s+\frac{1}{2}}}$$

$\Gamma(s+1) = s!$ for $s \in \mathbb{Z}$

$$= \frac{(s-\frac{1}{2})(s-\frac{3}{2}) \cdots \Gamma(\frac{1}{2})}{2a^{s+\frac{1}{2}}}$$

$\Gamma(s+1) = s\Gamma(s)$

$$= \frac{(2s-1)(2s-3) \cdots \sqrt{\pi}}{2^s \cdot 2a^s \cdot \sqrt{a}}$$

$\Gamma(\frac{1}{2}) = \sqrt{\pi}$

$$= \frac{(2s-1)!!}{2^{s+1} a^s} \sqrt{\frac{\pi}{a}}$$

13.1.21 For the gamma distribution

$$f(x) = \begin{cases} \dfrac{1}{\beta^{\alpha}\Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0, \\ 0, & x \le 0, \end{cases}$$

show that

(a) $\langle x \rangle$, the mean value of x, is equal to $\alpha\beta$,

(b) σ^2, its variance, defined as $\langle x^2 \rangle - \langle x \rangle^2$, has the value $\alpha\beta^2$.

Goal & Find μ and σ^2 of the Gamma Distributions

Solution

Recall that the expectation of a f^{n} of a random variable is

$$\langle f(x) \rangle = \int_{-\infty}^{\infty} f(x) \cdot p_x(x)\, dx \qquad \text{where } p_x(x) \text{ is the p.d.f of } X$$

a) $$\langle x \rangle = \frac{1}{\beta^{\alpha}\Gamma(\alpha)} \int_0^{\infty} x \cdot x^{\alpha-1} e^{-x/\beta}\, dx$$

$$= \frac{1}{\beta^{\alpha}\Gamma(\alpha)} \int_0^{\infty} x^{\alpha} e^{-x/\beta}\, dx$$

Integration by Sub. $\quad u = x/\beta \quad du = dx/\beta$

$$= \frac{1}{\beta^{\alpha}\Gamma(\alpha)} \int_0^{\infty} \beta^{\alpha} u^{\alpha} e^{-u} \beta\, du$$

$$= \frac{1}{\beta^{\alpha}\Gamma(\alpha)} \int_0^{\infty} \beta^{\alpha} u^{\alpha} e^{-u} \beta\, du$$

$$= \beta \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} = \alpha\beta$$

b) We find $\langle x^2 \rangle$. Then, construct σ^2.

$$\langle x^2 \rangle = \frac{1}{\beta^{\alpha}\Gamma(\alpha)} \int_0^{\infty} x^{\alpha+1} e^{-x/\beta}\, dx$$

$$= \frac{1}{\beta^{\alpha}\Gamma(\alpha)} \int_0^{\infty} \beta^{\alpha+1} u^{\alpha+1} e^{-u} \beta\, du$$

Integration by Sub. $\quad u = x/\beta \quad du = dx/\beta$

$$= \beta^2 \frac{\Gamma(\alpha+2)}{\Gamma(\alpha)} = (\alpha+1)\alpha\beta^2$$

Writing down σ^2

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2$$

$$= \alpha(\alpha+1)\beta^2 - \alpha^2\beta^2$$

$$= \alpha\beta^2$$ ▨

13.3.1 Verify the following beta function identities:

(a) $B(a,b) = B(a+1,b) + B(a,b+1)$,

(b) $B(a,b) = \dfrac{a+b}{b} B(a,b+1)$,

(c) $B(a,b) = \dfrac{b-1}{a} B(a+1,b-1)$,

(d) $B(a,b)B(a+b,c) = B(b,c)B(a,b+c)$.

Recall :

$$B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}\,dt$$

a) $$B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}\,dt$$

$$= \int_0^1 t^{a-1}(1-t)^{b-1}\underbrace{(t+1-t)}_{1}\,dt$$

$$= \int_0^1 t^{a}(1-t)^{b-1}\,dt + \int_0^1 t^{a-1}(1-t)^{b}\,dt$$

$$= B(a+1,b) + B(a,b+1)$$

b) $$B(a,b) = \int_0^1 \underbrace{t^{a-1}}_{u}\,\underbrace{(1-t)^{b-1}\,dt}_{dv}$$

$$= t^{a-1}\cdot\frac{(1-t)^b}{b}\Big|_0^1 - \int_0^1 \frac{(1-t)^b}{b}\cdot(a-1)t^{a-2}\,dt$$

$$= \frac{a-1}{b}\,B(a-1,b+1) \qquad (1)$$

From ⓐ ,

$$B(a,b) = B(a+1,b) + B(a,b+1)$$

$$= \frac{a}{b}B(a,b+1) + B(a,b+1) \qquad (11)$$

$$= \frac{a+b}{b}B(a,b+1)$$

c) $B(a,b) = \int_0^1 \underbrace{t^{a-1}}_{dv} \underbrace{(1-t)^{b-1}}_{u} \underbrace{dt}_{dv}$

$= (1-t)^{b-1} \cdot \frac{t^a}{a}\Big|_0^1 - \int_0^1 \frac{t^a}{a} \cdot -(b-1)(1-t)^{b-2}\, dt$

$= \frac{b-1}{a} B(a+1, b-1)$

d) $B(a,b)\, B(a+b,c) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \cdot \frac{\Gamma(a+b)\Gamma(c)}{\Gamma(a+b+c)} \cdot \frac{\Gamma(b+c)}{\Gamma(b+c)}$

$= \frac{\Gamma(b)\Gamma(c)}{\Gamma(b+c)} \cdot \frac{\Gamma(a)\Gamma(b+c)}{\Gamma(a+b+c)}$

$= B(b,c) \cdot B(a, b+c)$

Using $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$ would solve these in a pinch!

13.3.16 The beta distribution of probability theory has a probability density

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\beta)}\, x^{\alpha-1}(1-x)^{\beta-1},$$

with x restricted to the interval (0, 1). Show that

(a) $\langle x\rangle$, the mean value, is $\dfrac{\alpha}{\alpha+\beta}$.

(b) σ^2, its variance, is $\langle x^2\rangle - \langle x\rangle^2 = \dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.

Goal : Find μ and σ^2 of the Beta Distribution

Solution

Recall that the expectation of a f^{n} of a random variable is

$$\langle f(x)\rangle = \int_{-\infty}^{\infty} f(x)\cdot p_x(x)\,dx \qquad \text{where } p_x(x) \text{ is the p.d.f of } X$$

a) $$\langle x\rangle = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\beta)} \int_0^1 x\cdot x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\beta)}\cdot B(\alpha+1,\beta)$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\beta)}\cdot \frac{\alpha\Gamma(\alpha)\;\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}$$

$$= \frac{\alpha}{\alpha+\beta}$$

b) We find $\langle x^2\rangle$. Then, construct σ^2.

$$\langle x^2\rangle = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\beta)} \int_0^1 x^2\cdot x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\beta)}\cdot B(\alpha+2,\beta)$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\beta)}\cdot \frac{(\alpha+1)\,\alpha\Gamma(\alpha)\;\Gamma(\beta)}{\Gamma(\alpha+\beta+2)}$$

$$= \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}$$

Writing down σ^2

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2$$

$$= \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \frac{\alpha^2}{(\alpha+\beta)^2}$$

$$= \frac{\alpha(\alpha+1)(\alpha+\beta) - \alpha^2(\alpha+\beta+1)}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$= \frac{\alpha^2 + \alpha\beta - \alpha^2}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$= \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

15.1.2 Derive the following closed formula for the Legendre polynomials $P_n(x)$.

$$P_n(x) = \sum_{k=0}^{[n/2]} (-1)^k \frac{(2n-2k)!}{2^n k!\,(n-k)!\,(n-2k)!}\, x^{n-2k},$$

where $[n/2]$ stands for the integer part of $n/2$.

Hint. Further expand Eq. (15.12) and rearrange the resulting double sum.

Goal : Derive the closed formula for $P_n(x)$

Solution

The generating function of $P_n(x)$ is :

$$g(x,t) = (1-2xt+t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x) \cdot t^n$$

Using binomial expansion of the generating function $g(x,t)$:

$$(1-2xt+t^2)^{-1/2} = \sum_{n=0}^{\infty} \binom{-1/2}{n} (-2xt+t^2)^n$$

Working with each term seperately :

Recall : $\binom{n}{k} = \frac{n(n-1)\cdots(n-(k-1))}{k!}$

So, $\binom{-1/2}{n} = \frac{\overbrace{(-1/2)(-3/2)\cdots(-1/2-(n-1))}^{n \text{ terms}}}{n!}$

$$= \frac{(-1)^n \cdot (1\cdot 3\cdot 5\cdots 2n-1)}{2^n\, n!} \times \frac{2\cdot 4\cdot 6\cdots 2n}{2^n\, n!}$$

$$= \frac{(-1)^n\,(2n)!}{(2^n\, n!)^2}$$

For the n-power term, we use the binomial expansion :

$$(-2xt+t^2)^n = \sum_{k=0}^{n} \binom{n}{k} (-2xt)^{n-k} \cdot (t^2)^k$$

$$= \sum_{k=0}^{n} \frac{n!}{k!\,(n-k)!} (-1)^{n-k} \cdot 2^{n-k} \cdot x^{n-k} \cdot t^{n+k}$$

Putting the term back into $g(x,t)$ &

$$g(x,t) = \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{(-1)^n (2n)!}{(2^n n!)^2} \frac{n!}{k!(n-k)!} (-1)^{n-k} \cdot 2^{n-k} \cdot x^{n-k} \cdot t^{n+k}$$

$$= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{(-1)^k (2n)!}{2^{n+k} n! k! (n-k)!} x^{n-k} t^{n+k}$$

Almost there! Ideally, we would reorder indices so that $n+k \to n$ somehow.
This is to finally get $P_n(x)$.

We can do that by introducing $m = n + k$

Let's carefully consider the change on the summation indices

Before Index Change

$0 \leq k \leq n$

$0 \leq n$

After Index Change

$0 \leq k \leq m-k \to 0 \leq k \leq \lfloor \frac{m}{2} \rfloor$ $\quad 2k \leq m$

$0 \leq m-k \longrightarrow 0 \leq m$ $\quad k_{min} = 0$

floor f^n ceiling f^n

Why do we use $\lfloor \rfloor$ not $\lceil \rceil$?

If m even, $k \leq m/2$ & $m/2$ is an integer so $m/2$ is attained

If m odd, $k \leq m/2$ & $m/2$ is rational so k_{max} is the integer before $\frac{m}{2}$ OR $\lfloor \frac{m}{2} \rfloor$

Phew! With a change of indices &

$$g(x,t) = \sum_{m=0}^{\infty} \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{(-1)^k (2m-2k)!}{2^m (m-k)! k! (m-2k)!} x^{m-2k} t^m$$

$P_n(x)$ by def^n of generating f^n

15.4.3 Prove that

$$P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x),$$

where $P_l^m(x)$ is defined by

$$P_l^m(x) = \frac{(-1)^m}{2^l l!}(1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}}(x^2-1)^l.$$

Hint. One approach is to apply Leibniz's formula to $(x+1)^l(x-1)^l$.

Goal: Prove that $P_l^{-m}(x)$ has the form above.

Solution

Using the hint above, we use Leibniz's formula on both $P_l^m(x)$ and $P_l^{-m}(x)$ removing the derivatives and trying to glean the differences.

Leibniz's formula: $\frac{d^n}{dx^n}[f(x)\,g(x)] = \sum_{k=0}^{n} \binom{n}{k} \frac{d^k}{dx^k} f(x) \frac{d^{n-k}}{dx^{n-k}} g(x)$

The derivatives are of most bother. Let's investigate $\frac{d^{l-m}}{dx^{l-m}}(x^2-1)^l$ first.

$$\frac{d^{l-m}}{dx^{l-m}}(x-1)^l(x+1)^l = \sum_{k=0}^{l-m} \binom{l-m}{k} \underbrace{\frac{d^k}{dx^k}(x-1)^l}_{D_1} \underbrace{\frac{d^{l-m-k}}{dx^{l-m-k}}(x+1)^l}_{D_2}$$

D_1: $\frac{d^k}{dx^k}(x-1)^l = \overbrace{l\cdot(l-1)\cdots(l-(k-1))}^{k \text{ terms}}(x-1)^{l-k}$

$$= \frac{l!}{(l-k)!}(x-1)^{l-k}$$

D_2: $\frac{d^{l-m-k}}{dx^{l-m-k}}(x+1)^l = \frac{l!}{(l-[l-m-k])!}(x+1)^{l-(l-m-k)}$ using D_1

$$= \frac{l!}{(m+k)!}(x+1)^{m+k}$$

The identities above are more than enough to rid $P_l^m(x)$ & $P_l^{-m}(x)$ of the derivatives!

$$P_\ell^{-m}(x) = \frac{(-1)^{-m}}{2^\ell \ell!}(1-x^2)^{-m/2} \sum_{k=0}^{\ell-m} \binom{\ell-m}{k} \frac{\ell!}{(\ell-k)!}(x-1)^{\ell-k} \frac{\ell!}{(m+k)!}(x+1)^{m+k}$$

As for $P_\ell^m(x)$ &

$$P_\ell^m(x) = \frac{(-1)^m}{2^\ell \ell!}(1-x^2)^{m/2} \sum_{k=0}^{\ell+m} \binom{\ell+m}{k} \frac{d^k}{dx^k}(x-1)^\ell \frac{d^{\ell+m-k}}{dx^{\ell+m-k}}(x+1)^\ell$$

Hmm... Our derivates may exceed ℓ times yielding zero terms. We can remove those from our sum

Any values of $k<m$ and $k>\ell$ lead to zero terms. Therefore,

$$P_\ell^m(x) = \frac{(-1)^m}{2^\ell \ell!}(1-x^2)^{m/2} \sum_{k=m}^{\ell} \binom{\ell+m}{k} \frac{d^k}{dx^k}(x-1)^\ell \frac{d^{\ell+m-k}}{dx^{\ell+m-k}}(x+1)^\ell$$

Using Identity D, & $\frac{d^k}{dx^k}x^\ell = \frac{\ell!}{(\ell-k)!}x^{\ell-k}$

$$P_\ell^m(x) = \frac{(-1)^m}{2^\ell \ell!}(1-x^2)^{m/2} \sum_{k=m}^{\ell} \binom{\ell+m}{k} \frac{\ell!}{(\ell-k)!}(x-1)^{\ell-k} \cdot \frac{\ell!}{(k-m)!}(x+1)^{k-m}$$

$$= \frac{(-1)^m}{2^\ell \ell!}(1-x^2)^{m/2} \sum_{k=0}^{\ell-m} \binom{\ell+m}{k+m} \frac{\ell!}{(\ell-k-m)!}(x-1)^{\ell-k-m} \frac{\ell!}{k!}(x+1)^k$$

Index change
$k = k' - m$

$$= \frac{(-1)^m (-1)^m}{2^\ell \ell!}(1-x^2)^{-m/2} \sum_{k=0}^{\ell-m} \binom{\ell+m}{k+m} \frac{\ell!}{(\ell-k-m)!}(x-1)^{\ell-k} \frac{\ell!}{k!}(x+1)^{k+m}$$

$(1-x^2)^m = (-1)^m(x-1)^m(x+1)^m$

$$= (-1)^m \frac{(-1)^m}{2^\ell \ell!}(1-x^2)^{-m/2} \sum_{k=0}^{\ell-m} \frac{(\ell+m)!}{(\ell-m)!} \binom{\ell-m}{k} \frac{\ell!}{(\ell-k)!}(x-1)^{\ell-k} \frac{\ell!}{(m+k)!}(x+1)^{m+k}$$

Using Identity I next page

$$= (-1)^m \frac{(\ell+m)!}{(\ell-m)!} P_\ell^{-m}(x)$$

▨

Identity I 8

$$\binom{\ell+m}{k+m}\frac{1}{(\ell-k-m)!}\frac{1}{k!} = \frac{(\ell+m)!}{(k+m)!(\ell-k)!}\frac{1}{(\ell-k-m)!}\frac{1}{k!}$$

$$= \frac{(\ell+m)!}{(\ell-m)!}\underbrace{\frac{(\ell-m)!}{(\ell-k-m)!\,k!}}_{\binom{\ell-m}{k}}\cdot\frac{1}{(k+m)!(\ell-k)!}$$